

EdiGlobe VLSI Internship

Minor Project

Project Title: Realize JKFF using DFF and write the verilog code for a JK Flip flop, and its testbench. Use all possible combinations of inputs to test its working

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Software used: EDA playground

1. Objective

To design and implement a JK Flip-Flop using a D Flip-Flop in Verilog HDL and verify its functionality by simulating all possible input combinations using a testbench in EDA Playground.

2. Theory

A JK Flip-Flop is a sequential logic circuit that removes the invalid state of the SR Flip-Flop. A JK Flip-Flop can be implemented using a D Flip-Flop by generating the D input from the characteristic equation.

3. Characteristic Equation

$$D = J\bar{Q} + \bar{K}Q$$

4. Truth Table

J K	Operation	Next State
00	Hold	Q
01	Reset	0
10	Set	1
11	Toggle	$\bar{Q}$

5. Verilog Design Code

```
`timescale 1ns/1ps

module d_flipflop(
    input d,
    input clk,
    output reg q
);
always @(posedge clk)
begin
    q <= d;
end
endmodule

module jk_flipflop(
    input j,
    input k,
    input clk,
    output q
);
wire d;
wire q_internal;
assign d = (j & ~q_internal) | (~k & q_internal);
d_flipflop ff1(
    .d(d),
    .clk(clk),
```

```
.q(q_internal)  
);  
assign q = q_internal;  
endmodule
```

6. Testbench Code

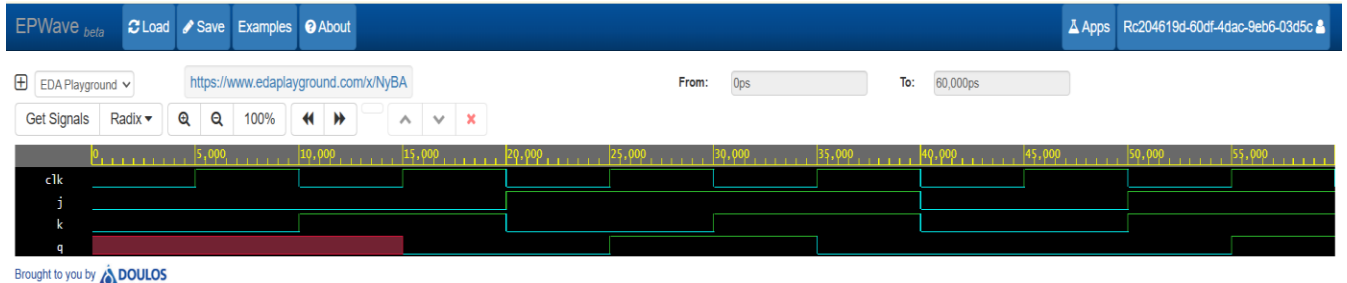
```
`timescale 1ns/1ps
module testbench;
reg j;
reg k;
reg clk;
wire q;
jk_flipflop uut(
    .j(j),
    .k(k),
    .clk(clk),
    .q(q)
);
always #5 clk = ~clk;
initial
begin
    $dumpfile("jk_flipflop.vcd");
    $dumpvars(0, testbench);
    clk = 0;
    $display("Time\tJ\tK\tQ");
    $monitor("%0t\t%b\t%b\t%b", $time, j, k, q);
    j = 0; k = 0;
    #10;
    j = 0; k = 1;
    #10;
```

```
j = 1; k = 0;  
#10;  
j = 1; k = 1;  
#10;  
j = 0; k = 0;  
#10;  
j = 1; k = 1;  
#10;  
$finish;  
end  
endmodule
```

7. Simulation Output

```
Log Share
[2026-07-14 06:34:15 UTC] iverilog '-wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out
VCD info: dumpfile jk_flipflop.vcd opened for output.
Time   J   K   Q
0      0   0   x
10000  0   1   x
15000  0   1   0
20000  1   0   0
25000  1   0   1
30000  1   1   1
35000  1   1   0
40000  0   0   0
50000  1   1   0
55000  1   1   1
testbench.sv:47: $finish called at 60000 (1ps)
Finding VCD file...
./jk_flipflop.vcd
[2026-07-14 06:34:16 UTC] Opening EPWave...
Done
```

8. Waveform



9. Result

The JK Flip-Flop was successfully realized using a D Flip-Flop. Simulation verified the Hold, Reset, Set and Toggle operations for all possible input combinations.

10. Conclusion

The project demonstrates the implementation and verification of a JK Flip-Flop using Verilog HDL. The design compiled successfully and the simulation results matched the expected behavior.